Chapter 9

IS-MP-FE

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GE with Nominal Frictions

- Baseline RBC models study GE with real variables
  - Minimal, if any, frictions
  - *at least until I get to make my own structural macro class*

- Now, we'll study GE with a New Keynesian model
  - Including pricing frictions
  - We can also study short vs long-run outcomes
  - RBC models tend to focus on short run

› Is "short run" hyphenated? As an adjective, yes. As a noun, usually no. Do I care on the exam? No. Am I going to forget in my slides? Definitely.

Connecting Markets

› Before we dive into that, let's take stock of our markets

1. Labor equilibrium: *N* and *W*
   - Nominal wage = real wage × Price level: $W = wP$
2. Loanable funds equilibrium: *S,I* and *i*
   - A quick link with Fisher: $i = r + E[\pi']$
3. Reserves equilibrium: Res. and *i*

The Nominal Interest Rate

- Two of those markets clear with *i*
  - Loanable funds
  - Reserves

- Start there and see if we can bring the labor market in later

Loanable Funds with *i*

› Previously, our y-axis was *r*

› We know $i = r + E[\pi']$

- With adaptive expectations, $E[\pi']$ takes time to update
- *In the short run, a change in r is the same as a change in i*

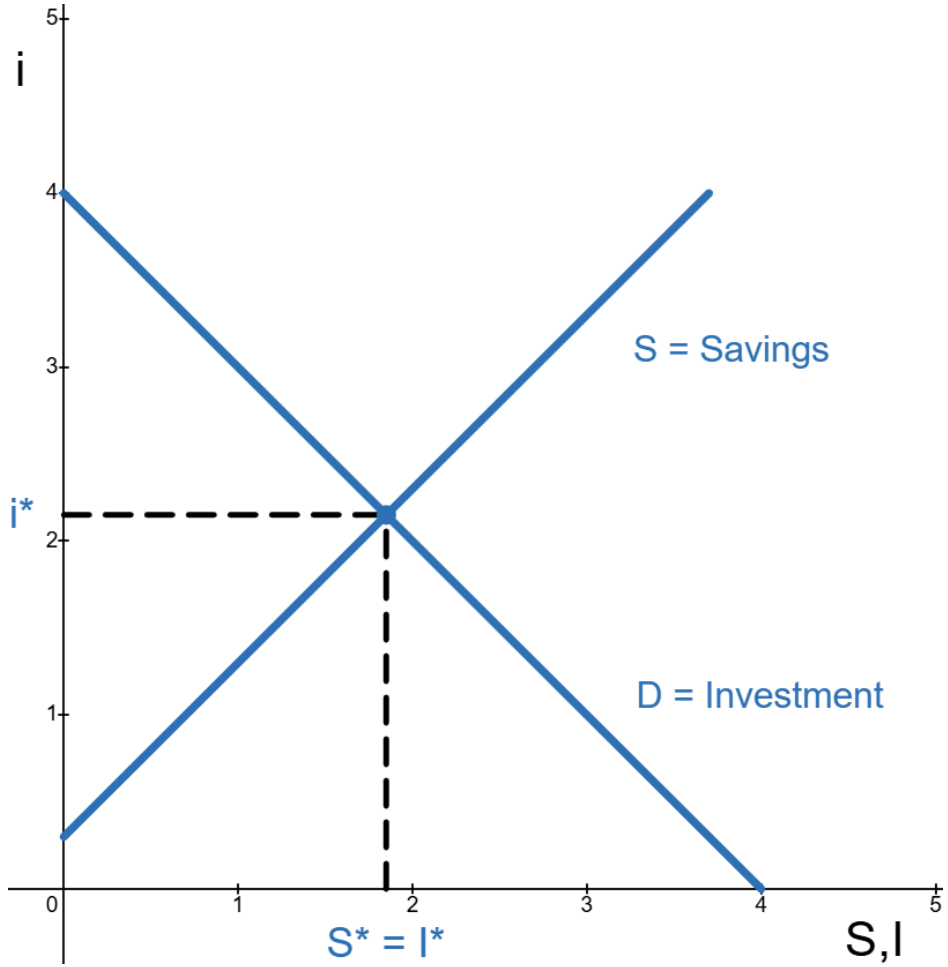
Previously, on Intermediate Macro…

› Closed economy equilibrium: $S = I$

› What happens if we change Y, *all else equal*…

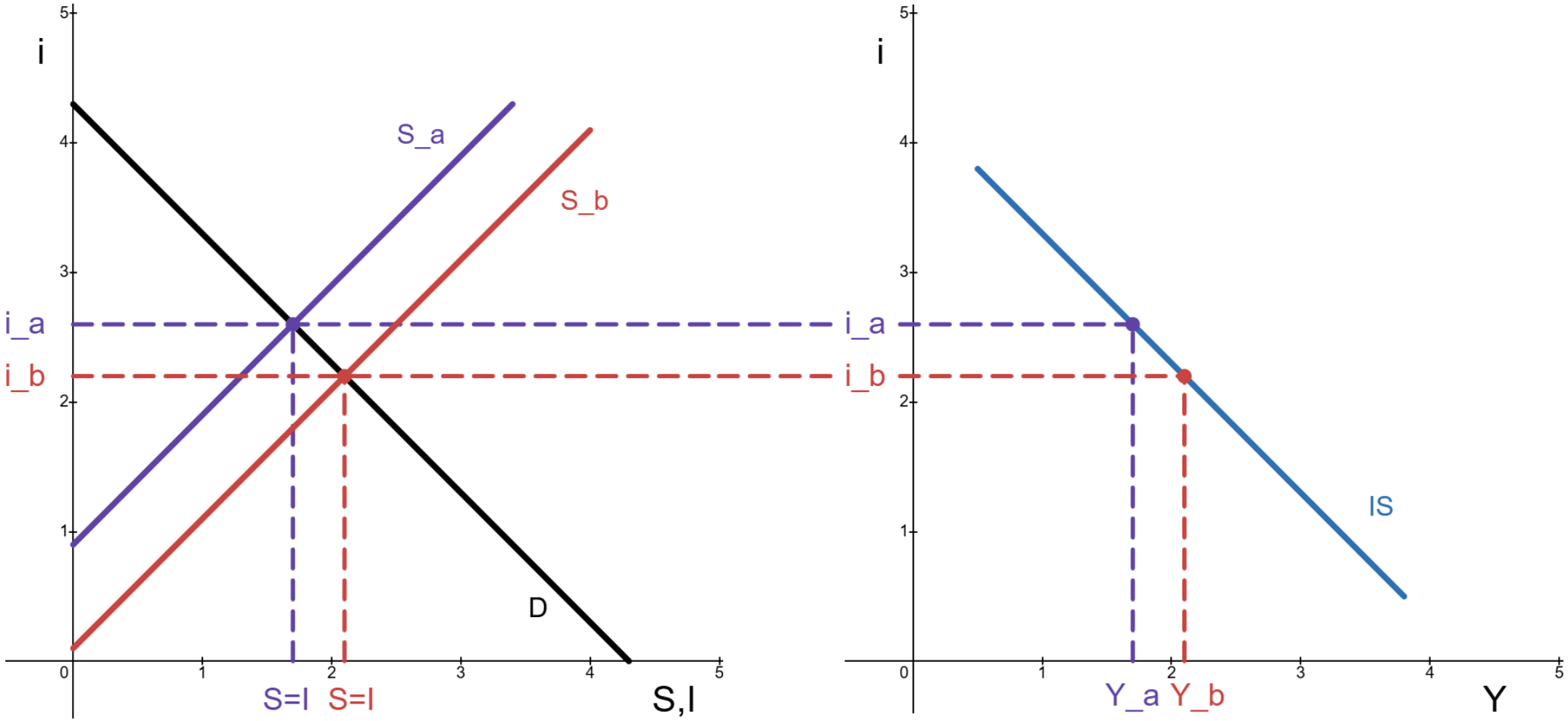
Increase in *Y*

- Positive WE
  - *c* and *c′* increase
  - Strict convexity
- More *c′* means some of that extra *Y* must be saved
- Leads to a new equilibrium



Remember: E[π'] fixed at first

Investment-Savings

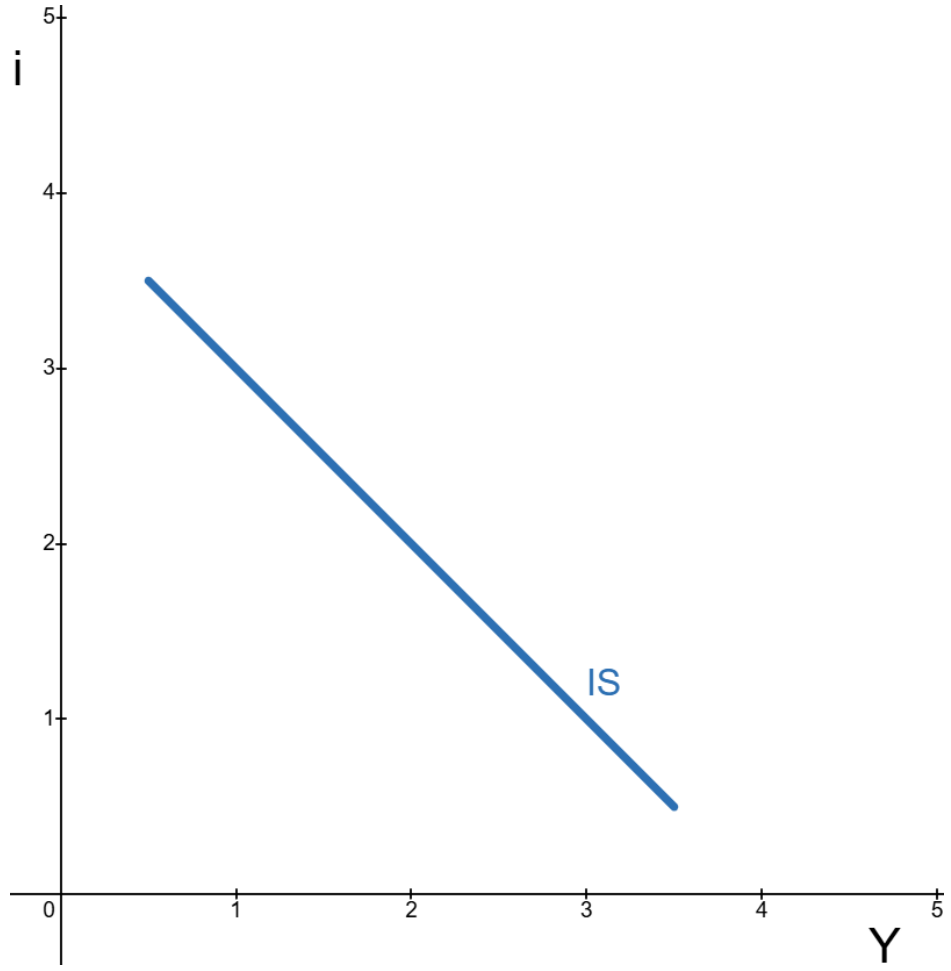


The Investment-Savings Curve

- The **IS Curve** plots the different combinations of *Y* and *i* where the goods market is in equilibrium
  - For a given *Y*, there is an *i* that clears the goods market

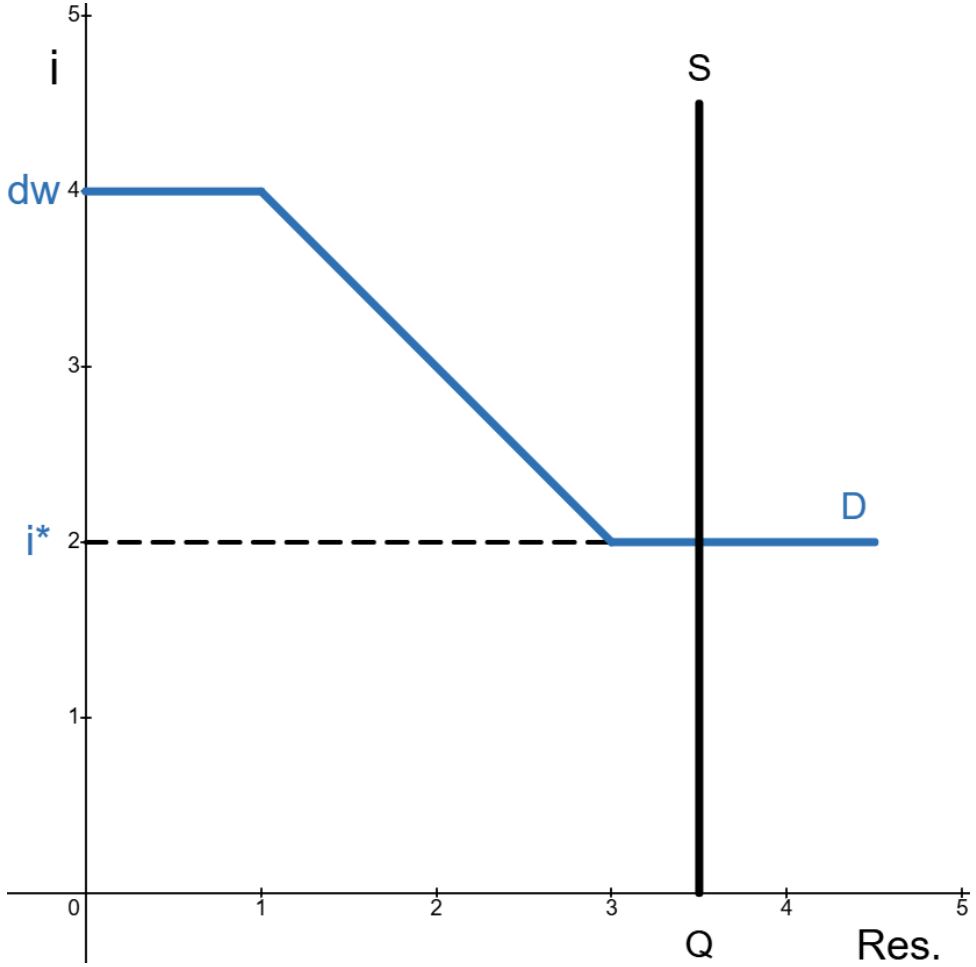
- *Anything that changes the equilibrium in the goods market will shift the IS curve*
  - Besides *Y being the initial change*, of course

The IS Curve

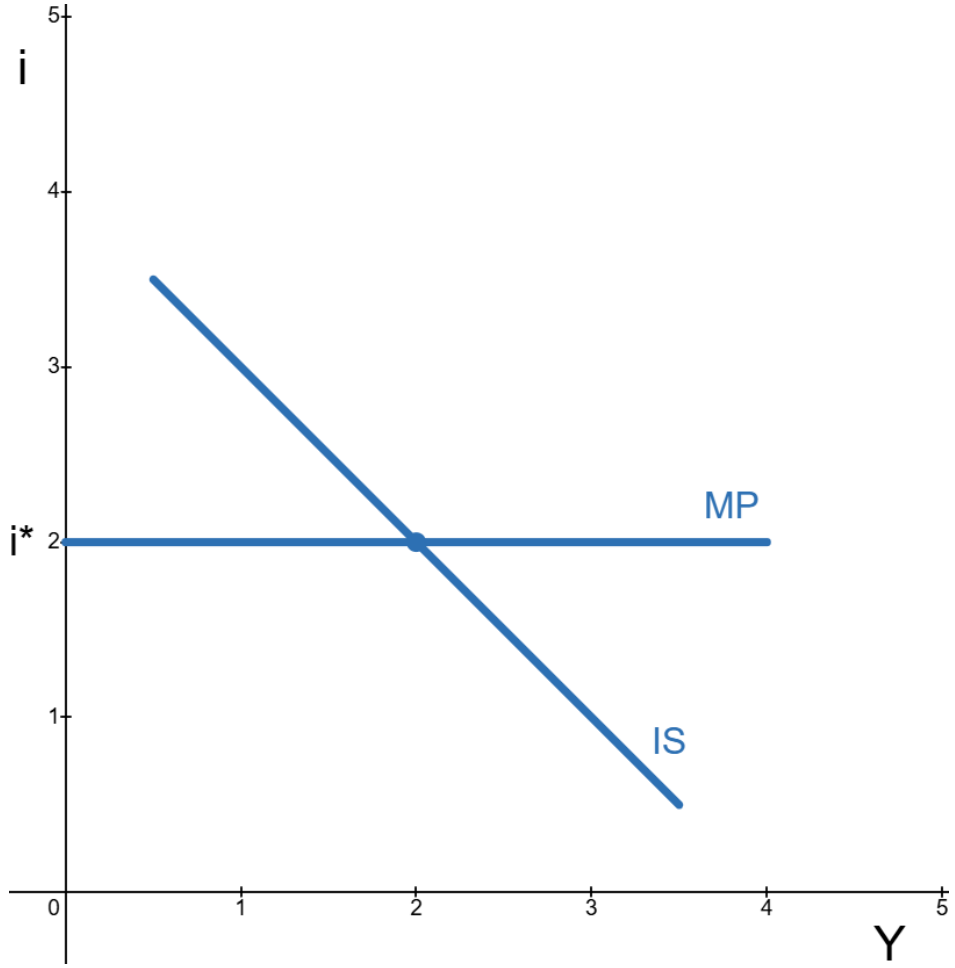


Adding the Reserves Market

- The Fed seems set on maintaining ample reserves
  - We'll use that assumption
- Fed can pick an *i*
- This *i* is our MP "curve"
  - Monetary Policy curve



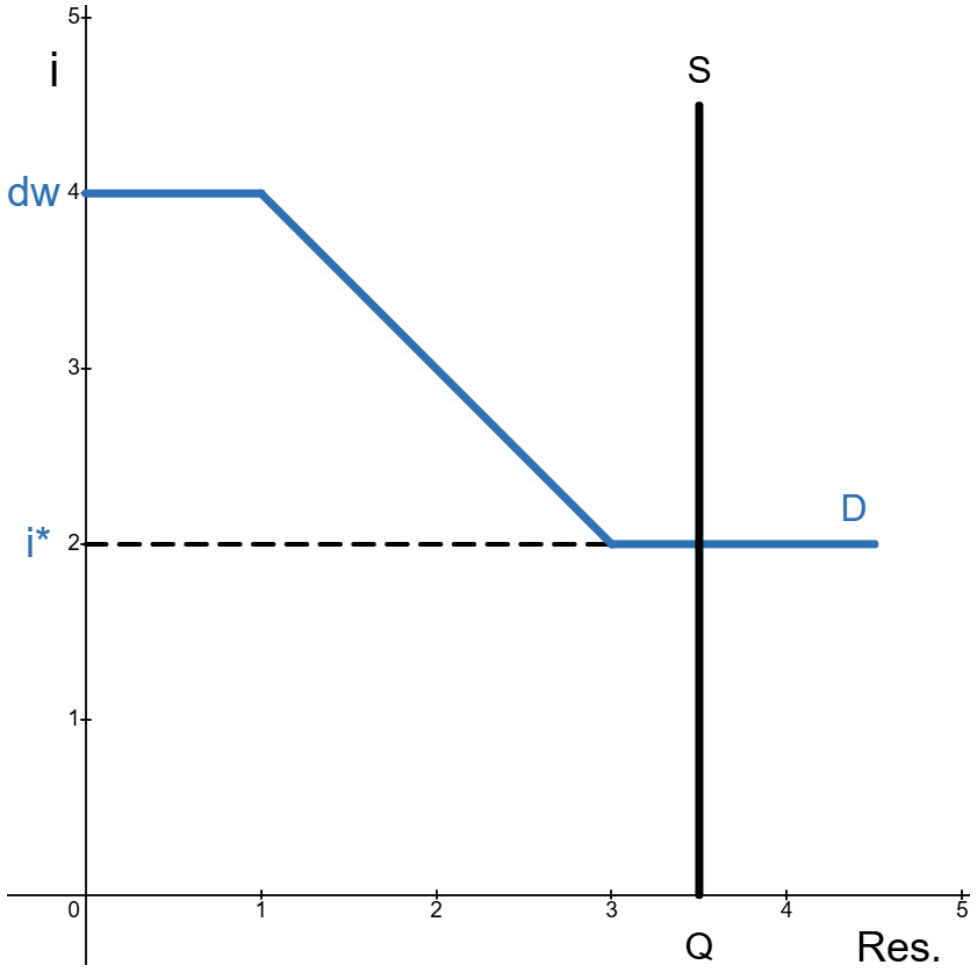
The IS-MP Model

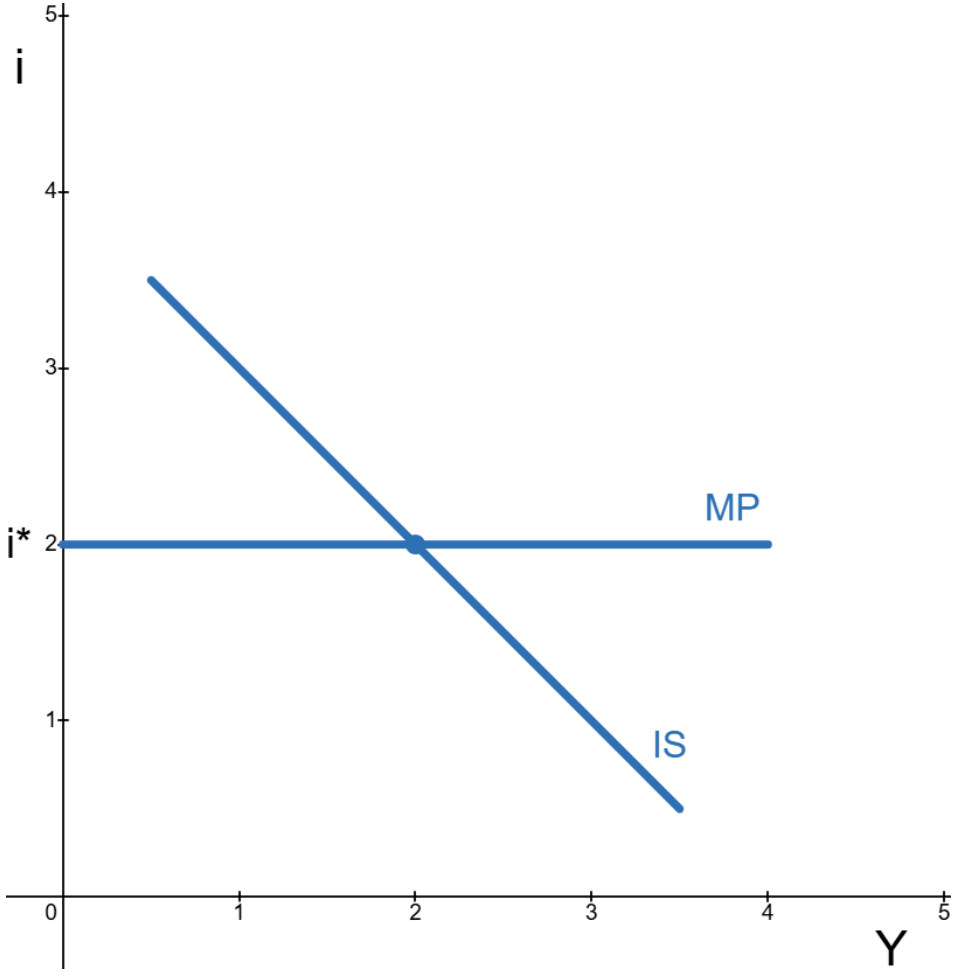


Shifts in the IS Curve: Increase in G

Shifts in the IS Curve: Increase in *MPK'*

A Fall in the IORB





Incorporating the Labor Market

- The final market is our labor market
  - Wage clears the labor market, not interest rates
  - How do we incorporate it?

- With our “full employment” definition from Ch. 4

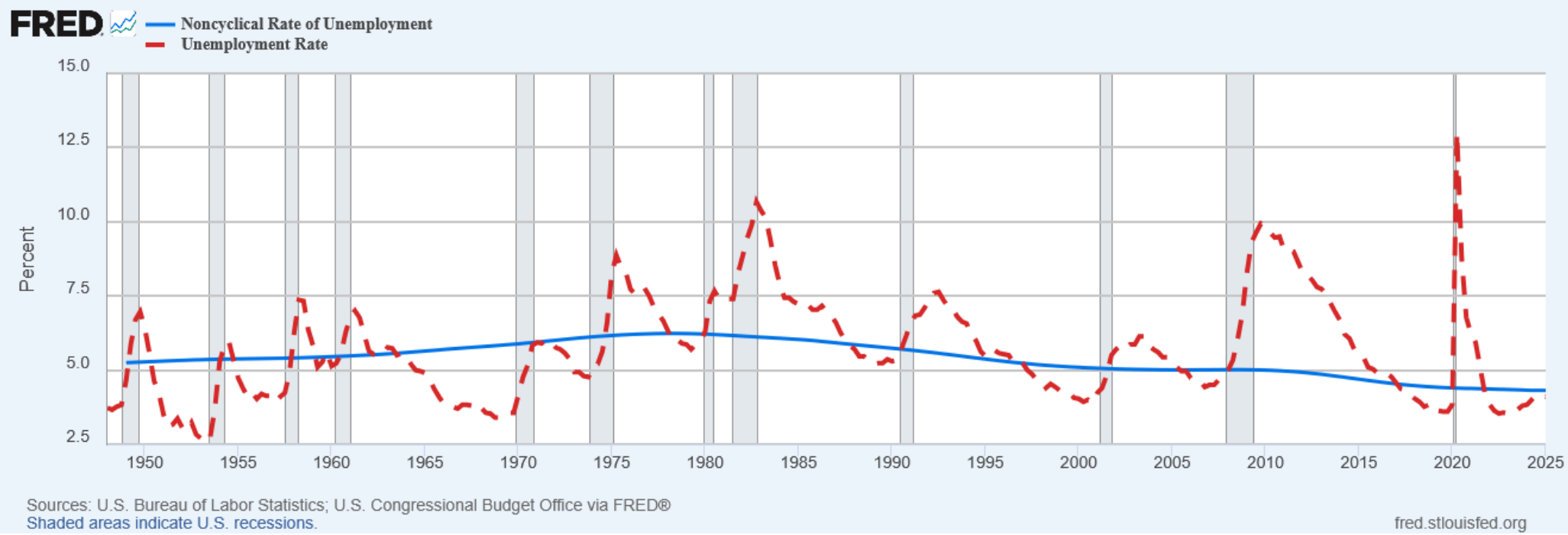
Returning to the Labor Market

- Labor market equilibrium implies *full employment* (*FE*)
  - Cyclical unemployment = 0

- Define this level of employment as $\bar{N}$
  - $\bar{Y} = zF(K, \bar{N})$

- Determinants of $\bar{N}$:
  - z
  - K
  - Structure of HH utility

Noncyclical Rate of Unemployment



Output Potential and *i*

- Since $\bar{N}$ doesn't move with *i*, we can capture it with $\bar{Y}$
  - Already have an axis with *Y*, so $\bar{Y}$ is just a specific point

- This point is **output potential**
  - Where the economy will end up in absence of shocks
  - Provides an anchor for the Fed to analyze economy

- Can separate the long run from short run to analyze the effects of needing time to reach our potential
  - Though, this does cause some linguistic tension

Long Run vs Next Period

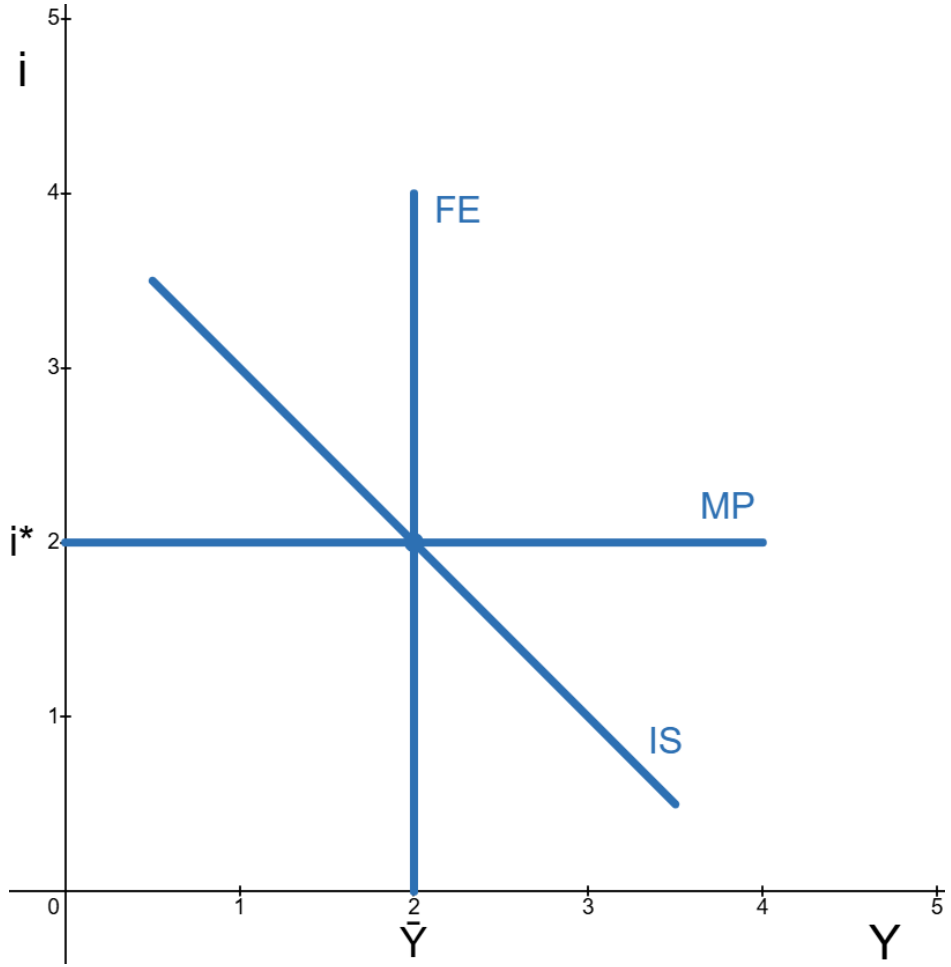
- Does moving to the long run mean we're looking at c' instead of c?
  - No. Next period variables have a clear time definition depending on what we study
    - Years, quarters, months, etc.

- The long run is abstract
  - However long it takes these frictions we've mentioned to "unstick"
  - Varies by firm, industry, etc.
  - More soon!

General Equilibrium

› What happens when the whole economy is in equilibrium?
  - $S = I$
  - At i^* set by the Fed
  - At full employment

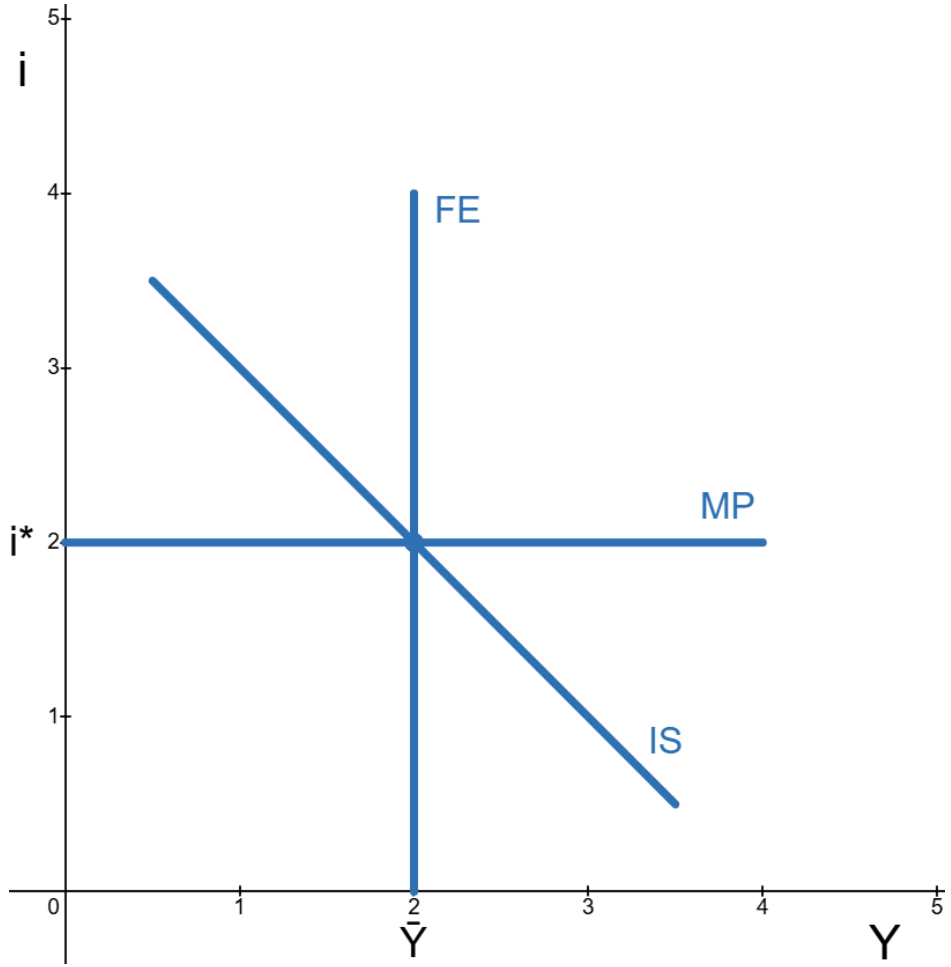
IS–MP–FE



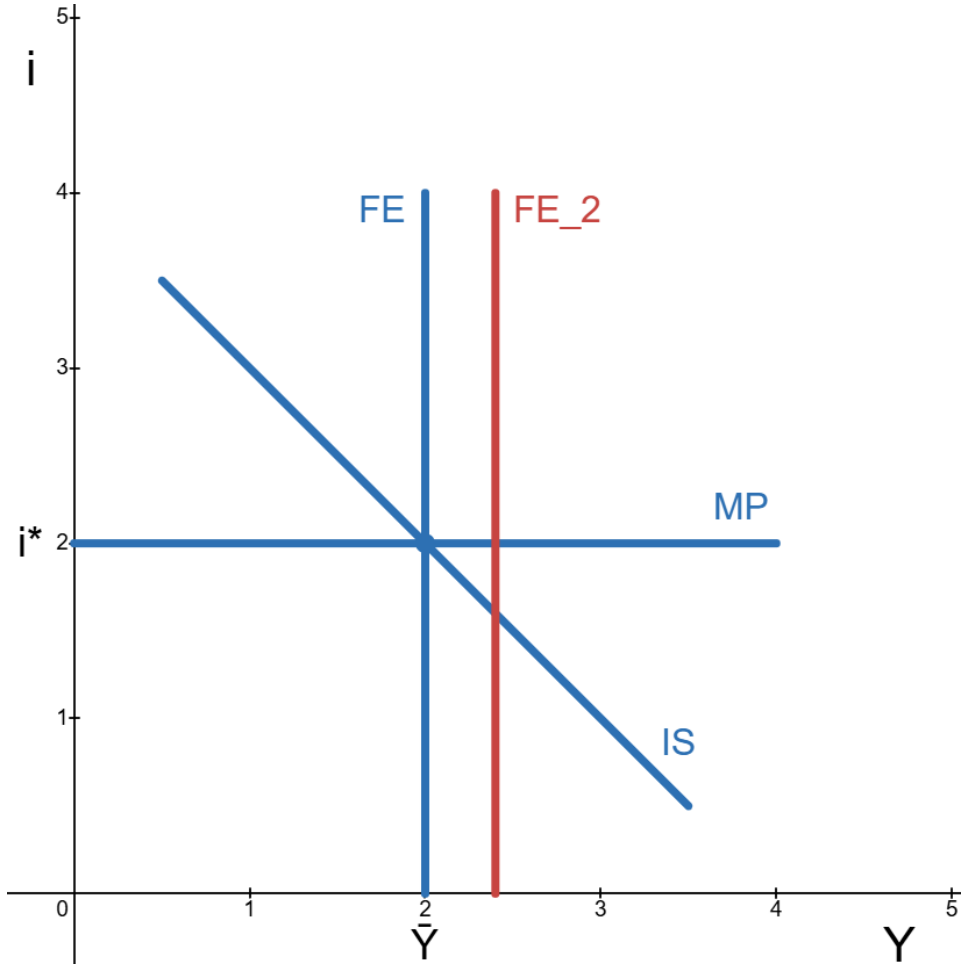
Shocks to TFP

- A *temporary* increase in TFP is in increase in z
  - Both K and N are more productive
  - $\bar{Y}$ increases
- An increase in *future* TFP is an increase in z'
- A *permanent* increase in TFP is in increase in z and z'
  - Beyond the bounds of this class

A *Temporary* Increase in TFP



Great. So, Where Are We?



The Output Gap & the Fed

› There is now an **output gap** in the economy
  - $Y - \overline{Y} \neq 0$

› The Fed can:
  - Do nothing; let the shock fade
  - Conduct monetary policy to close the gap

› Next chapter looks at short and long-run tradeoffs of each

References

1. U.S. Bureau of Labor Statistics, Unemployment Rate [UNRATE], retrieved from FRED, Federal Reserve Bank of St. Louis; https://fred.stlouisfed.org/series/UNRATE, .

2. U.S. Congressional Budget Office, Noncyclical Rate of Unemployment [NROU], retrieved from FRED, Federal Reserve Bank of St. Louis; https://fred.stlouisfed.org/series/NROU, .

A. All non-cited graphs for this course made by me on Desmos and MS Paint.

APPENDIX

IS–LM–FE (old)

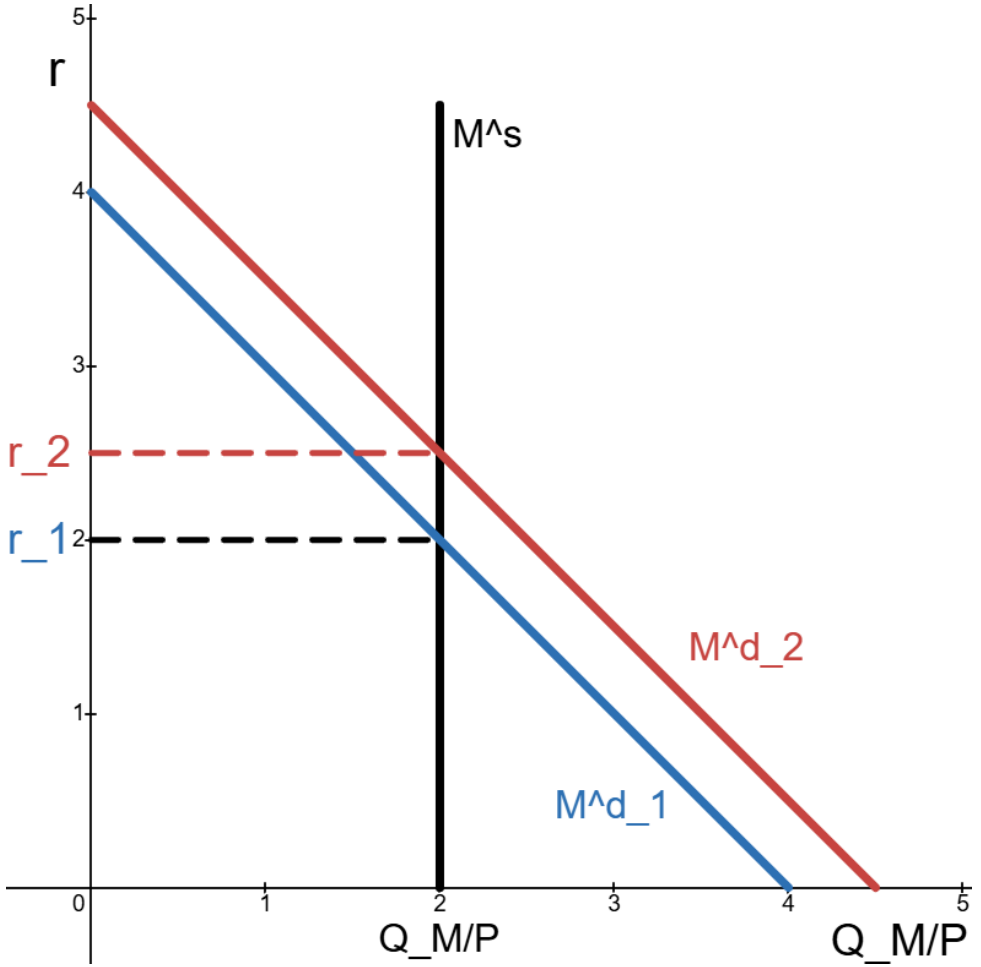
- Before the ample reserves system, monetary policy was concerned with moving M^s in the money market

- Same way we build the IS curve is applied to the LM curve
  - Liquidity Preference-Money Supply curve

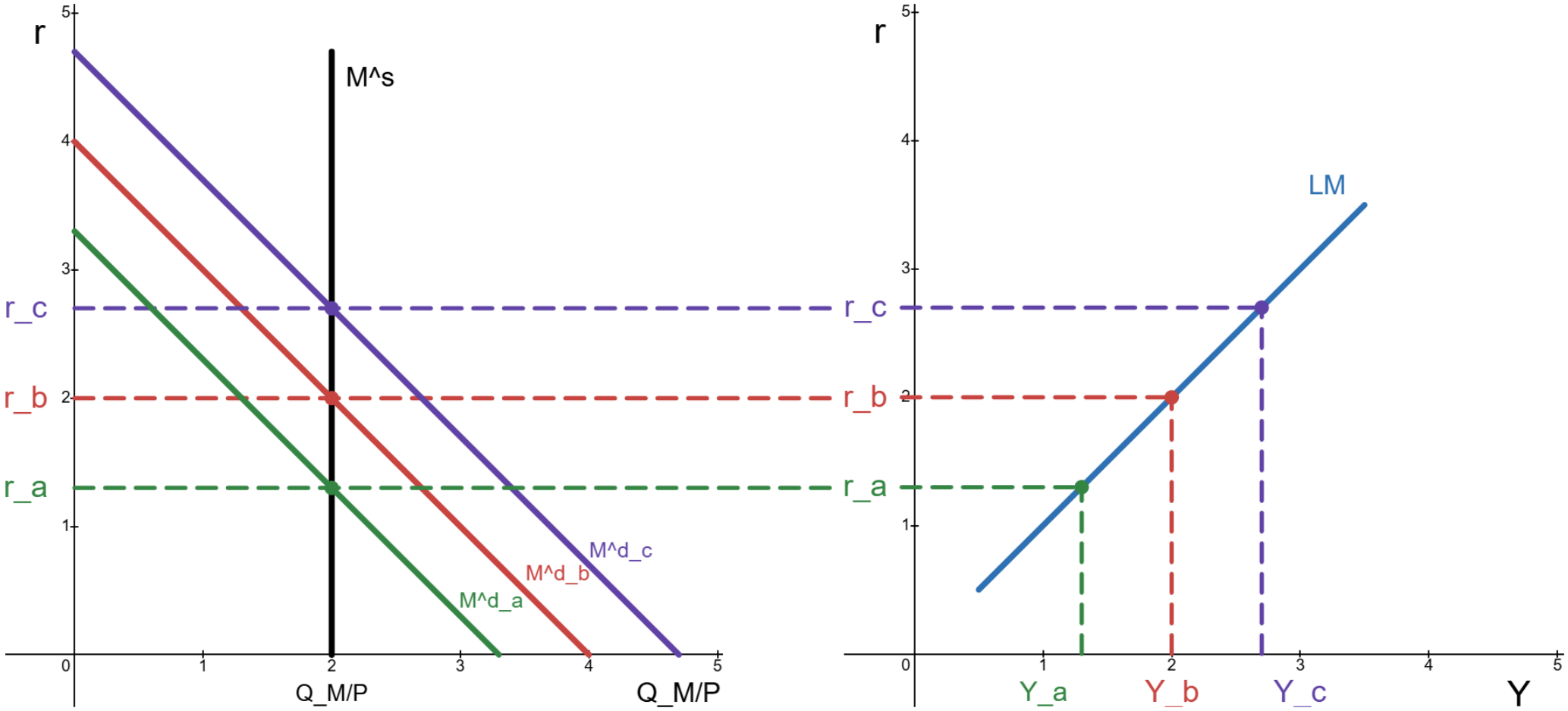
- Since the Fed isn't setting a nominal interest rate, we just use the real interest rate, r
  - Depends on P, since real money balances are: $\frac{M^s}{P}$
  - Still used in majority of macro classes, though I'm abandoning it for the modern analysis

Positive Relationship b/t *Y* and *r*

- If *Y* increases, so does *r*
- In fact, we have a new equilibrium when there is an increase in *Y*
- We can plot the equilibrium combinations of *Y* and *r*…



Liquidity Preference-Money Supply (LM)



The LM Curve

- The **LM curve** plots the different combinations of *Y* and *r* where the money market is in equilibrium
  - For a given *Y*, there is an *r* that clears the money market

- *Anything that changes the equilibrium in the money market will shift the LM curve*
  - Besides *Y being the initial change*, of course

IS–LM–FE

